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Nemotron-Labs-Diffusion

NVIDIA Unifies Autoregressive and Diffusion Language Modeling Into a Single Self-Speculating Architecture That Eliminates the Need for a Separate Draft Model

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Modern speculative decoding relies on a separate draft model to propose candidate tokens that a larger verifier model then accepts or rejects. Maintaining two models doubles deployment cost and complicates serving infrastructure. Nemotron-Labs-Diffusion eliminates the draft model entirely by training a single set of weights to operate in three modes: standard autoregressive (AR) decoding, parallel diffusion decoding, and self-speculation, where the diffusion head drafts while the AR head verifies—from the same checkpoint, with no weight switching.

AT A GLANCE

Problem: Speculative decoding requires a separate draft model, doubling parameter budget and serving complexity.

Why it matters: Throughput is the primary cost driver for large-scale LLM deployment; a 4-6× improvement directly reduces inference cost.

Method: Joint AR-diffusion training objective; at inference, the diffusion head proposes draft token blocks and the AR head verifies, all within one set of weights.

Result: 6.82 accepted tokens per step (vs. Eagle3’s 2.75); 4× higher throughput than Qwen3-8B on SPEED-Bench.

Evidence: Evaluated on SPEED-Bench, MT-Bench, MMLU, and HumanEval across 3B, 8B, and 14B model sizes.

The Big Picture

The dominant paradigm for efficient large language model inference is speculative decoding: a small, cheap draft model proposes a block of candidate tokens, and the large model verifies the entire block in a single forward pass. When the acceptance rate is high, this delivers near-linear speedups. The problem is that the draft model is a separate artifact that must be trained, maintained, and co-deployed with the verifier, doubling operational complexity.

Multi-token prediction (MTP) sidesteps the separate model by training the main model to predict multiple future tokens simultaneously using auxiliary heads. But MTP heads predict tokens autoregressively from a fixed left-to-right prior, limiting their ability to propose coherent multi-token continuations.

Nemotron-Labs-Diffusion observes that the diffusion training objective—which reconstructs any masked subset of tokens conditioned on unmasked context—is exactly the capability needed for a high-quality internal drafter. A model trained jointly with AR and diffusion objectives learns to both predict the next token from the left and fill any arbitrary masked block, making its diffusion head a natural draft model and its AR head a natural verifier.

Why Existing Approaches Fall Short

Eagle3, the leading separate-draft speculative decoding system, achieves 2.75 accepted tokens per step. Its primary weakness is that the draft model must be trained separately and kept synchronized with the verifier model through costly co-training or periodic retraining. Every new verifier checkpoint requires a corresponding draft model update.

MTP-based approaches (as used in DeepSeek-V3) train auxiliary prediction heads on the verifier model itself but restrict each head to predicting a single additional token conditioned on prior outputs. The heads predict tokens serially rather than in parallel, and their training signal is weaker because they never reconstruct arbitrary masked subsets—they only predict a fixed number of positions ahead.

Standard diffusion LMs (as in MDLM and Plaid) have high generation quality in parallel mode but suffer from a quality-speed tradeoff: fewer diffusion steps produce lower quality, while more steps reduce the throughput advantage over AR models.

Core Mechanism

Nemotron-Labs-Diffusion resolves these tensions by exploiting the complementarity between AR and diffusion objectives. The AR objective provides strong left-to-right linguistic priors that make verification reliable; the diffusion objective provides parallel reconstruction capability that makes drafting both fast and coherent.

In self-speculation mode, the model executes a draft-verify cycle. The diffusion head, given the current context, denoises a block of B masked positions in parallel using one or two diffusion steps. The AR head then scores each proposed token in a single forward pass (matching the cost of verifying B tokens in standard speculative decoding). Tokens above the acceptance threshold are committed; the rest trigger a fresh diffusion draft.

Technical Formulation

The joint training loss is:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{AR}} + (1 - \alpha) \mathcal{L}_{\text{diff}}$$

where $\mathcal{L}_{\text{AR}}$ is the standard cross-entropy on the next token and $\mathcal{L}_{\text{diff}}$ is the masked token reconstruction loss at variable mask rates $\rho \sim \text{Beta}(a, b)$:

$$\mathcal{L}_{\text{diff}} = -\mathbb{E}_{\rho, \mathcal{M}} \sum_{i \in \mathcal{M}} \log p_{\theta}(x_i | x_{\setminus \mathcal{M}})$$

with $\mathcal{M}$ the random mask set of expected size ρn . At $\alpha = 0.5$, both objectives contribute equally; ablations confirm this is the Pareto-optimal balance.

Token acceptance in self-speculation mode follows the standard speculative decoding criterion: draft token x_d is accepted with probability $\min(1, p_{\text{AR}}(x_d)/p_D(x_d))$, preserving the AR distribution exactly.

Learning or Inference Procedure

1. Train with joint AR+diffusion loss using standard next-token and masked-span reconstruction objectives.
2. Fine-tune an additional lightweight MLP sampler on diffusion sampling trajectories to reduce required denoising steps to 1–2.
3. At inference, select mode based on deployment constraints: AR mode for minimum per-token latency; diffusion mode for maximum throughput at batch size 1; self-speculation for the best throughput-quality tradeoff at any batch size.
4. Self-speculation loop: draft $B = 8$ tokens via 1-step diffusion; score with AR head; commit accepted tokens; redraft rejected suffix.

What the Guarantee Says

The acceptance-based guarantee from standard speculative decoding carries over: accepted tokens are drawn exactly from the AR model’s distribution, so Nemotron-Labs-Diffusion-self-speculation produces outputs statistically identical to pure AR sampling. No quality degradation is introduced by the self-speculation mechanism.

The throughput guarantee is empirical: on SPEED-Bench with SGLang on NVIDIA GB200, Nemotron-Labs-Diffusion-8B achieves $4\times$ higher throughput than Qwen3-8B at equivalent quality (within 1% on MT-Bench, MMLU, and HumanEval).

Experimental Findings

System	Accepted toks/step	Throughput
Qwen3-8B (AR only)	1.00	1.0×
DeepSeek-V3 MTP	2.11	1.9×
Eagle3 (separate draft)	2.75	2.6×
Nemotron-8B (ours)	6.82	4.0×

Quality scores (MT-Bench / MMLU / HumanEval) for Nemotron-8B are within 1% of Qwen3-8B AR baseline across all benchmarks. The 3B model variant outperforms the 7B Eagle3 draft+8B verifier pair on throughput while using 5B fewer total parameters.

Ablations and Interpretation

Removing diffusion training drops accepted tokens per step from 6.82 to 4.1, confirming the diffusion head’s centrality to draft quality. This is the most significant ablation result.

α sweep: Higher α (more AR weight) improves text quality metrics by up to 0.3% but reduces acceptance rate by up to 1.4 tokens per step. $\alpha = 0.5$ is the Pareto-optimal balance.

Mask rate schedule: Fixed mask rate $\rho = 0.15$ (like BERT) reduces accepted tokens per step by 1.2 relative to the Beta(1, 5) schedule, which samples mask rates across the full $[0, 1]$ range and builds more versatile reconstruction capability.

Self-speculation vs. MTP: At equal parameter budgets, self-speculation accepts 6.82 tokens per step versus MTP’s 2.11, because MTP heads can only predict a fixed number of positions while the diffusion head can fill any masked block with a single forward pass.

Reference

Yonggan Fu et al. **Nemotron-Labs-Diffusion: A Tri-Mode Language Model Unifying Autoregressive, Diffusion, and Self-Speculation Decoding**. arXiv:2607.05722, July 2026. <https://arxiv.org/abs/2607.05722>